Euclid, the Sulbasutras, and the Geometry of the Mind: A Mathematical Continuum from Antiquity to Brain-Machine Interfaces

Aman Chawla*

Across the vast temporal gulf separating ancient geometry from modern computational neuroscience lies an unbroken lineage of mathematical abstraction. Euclid's *Elements*, composed in the third century BCE, stands as the archetypal systematization of geometry, setting forth not only spatial truths but also the rigorous logical methodology that undergirds mathematical reasoning to this day. Yet Euclid was not alone in seeking the secrets of space. Even earlier, the Sulbasutras of ancient India, composed between 800 and 200 BCE, presented geometric rules necessary for constructing ritual altars. These texts contain the earliest known statements of the Pythagorean theorem, along with algorithms for constructing squares, circles, and more complex shapes. While these early geometric treatises were deeply rooted in cosmology and ritual, they nevertheless reflect a shared intuition: that the structure of space can be captured by abstract reasoning, and that this reasoning enables both understanding and construction.

Fast forward to the 21st century, and we encounter a striking descendant of this tradition in the computational modeling of neural phenomena. In the paper *On Axon Interaction and Its Role in Neurological Networks* by Chawla, Morgera, and Snider, geometry becomes the vehicle for understanding an entirely different domain: the electrophysiological behavior of axon tracts in the human nervous system. Through a novel generalization of the cable equation, the authors derive a mathematical framework that incorporates both the physical parameters and the spatial configurations of axons, uncovering the subtle effects of ephaptic coupling—non-synaptic interactions mediated by extracellular fields. What is remarkable is not only the biological relevance of their results, but the deep structural echo they carry of Euclid's and the Sulbasutras' geometric insights. Here, too, relationships among lengths, angles, and proximities reveal fundamental truths, now about neuronal function rather than altar construction or ideal triangles.

The essence of Euclid's *Elements* lies in its axiomatic structure: a small set of definitions, postulates, and common notions gives rise to a vast landscape of theorems, all connected by strict logical inference. Similarly, the Sulbasutras—though not formalized in the Euclidean style—provided

*in collaboration with an LLM

constructive algorithms that were precise, reproducible, and general. Chawla et al. operate within this very paradigm. Beginning with axiomatic assumptions about axonal geometry and biophysics—cable theory, ionic current models, and tract topology—they derive new equations describing the propagation and interaction of action potentials. The introduction of angular inclination (θ_i) and inter-axonal distances (captured in a phenomenological matrix W_{ip}) into the governing equations is reminiscent of the Sulbasutra principle that precise proportions and spatial arrangements influence not just the form but the function of a structure. Where the ancient texts used cord constructions to derive square roots and transform shapes, this modern work uses numerical simulation to explore how geometry modulates communication between neurons.

What emerges is not just a technical improvement over existing models but a conceptual shift. The nervous system, in this formulation, becomes a geometric communication network. Ephaptic coupling is shown to depend on the orientation and proximity of axons in a tract—parameters that were previously neglected or idealized in many studies. The simulations demonstrate that compressive injuries altering axonal angles can sever functional connectivity, and that changes in spacing modulate the delay and success of action potential propagation. These effects are not just local but systemic. Much like the altar-construction algorithms of the Sulbasutras, where a single length miscalculation could ruin a ritual structure's efficacy, a small misalignment in axons can disrupt an entire neural pathway. In this sense, both ancient and modern approaches share a respect for the precision and power of geometry.

This geometrically enriched model not only improves our understanding of neural function but also has profound implications for applied neuroscience. One particularly promising domain is brain-machine interfacing. Current neuroprosthetics often rely on simplified assumptions about neural encoding and spatial structure, leading to coarse interactions and limited adaptability. But a model that explicitly includes ephaptic coupling opens the door to entirely new prosthetic architectures. Rather than passively detecting neural spikes, future devices might actively participate in the ephaptic dynamics of axon bundles. By replicating the spatial structure and electrical environment of native tracts, such interfaces could synchronize with biological signals in real time, enabling more natural and efficient integration. This is not merely an engineering refinement—it is a conceptual realignment, recognizing that spatial geometry is as central to brain function as it is to Euclidean or Sulba constructions.

Furthermore, the work of Chawla and colleagues also touches upon synchronization and information transmission—concepts that resonate with modern information theory but have ancient analogs. The Sulbasutras, in their geometric constructions, often specified ratios and alignments with an eye to maintaining harmony. Likewise, Euclid's propositions culminate in the understanding of proportion, similarity, and congruence. In the brain, as in ritual space or pure geometry, synchronization reflects coherence. The authors show that axons of different diameters can, through ephaptic coupling, achieve phase synchronization despite initial lags in conduction velocity. This insight points toward the possibility of using spatial geometry to engineer coherent information flow, even in heterogeneous systems. Such a discovery reframes the brain not as a messy biological substrate but as a rigorously tunable geometric machine.

In reflecting on this intellectual lineage—from the altar builders of Vedic India, through the axiomatic idealism of Alexandria, to the computational neurobiologists of today—it becomes clear that geometry has always been a means of translating complexity into clarity. The nervous system, long considered an inscrutable tangle, now yields to mathematical analysis in a manner directly analogous to how the ancients made sense of space. But whereas the Sulbasutras aimed at metaphysical order and Euclid at logical purity, today's geometries are animated by a third goal: utility. The implications for neuroprosthetics, for understanding diseases such as multiple sclerosis, and for building brain-like machines are vast. Each simulation and equation derived in this study is not just a mathematical artifact, but a potential blueprint for real-world intervention.

In conclusion, the work of Chawla, Morgera, and Snider stands as a remarkable modern echo of humanity's oldest mathematical aspirations. It is a direct descendant of the Sulbasutras' constructive pragmatism and Euclid's logical rigor, yet directed toward a domain those early geometers could scarcely have imagined: the electric geometry of the mind. Their generalized model of axon interaction, rooted in physical law and spatial reasoning, reveals the brain as a structure to be understood not only chemically or statistically, but geometrically. The space between axons, their orientation, and the coupling they produce are as much a source of function as are the synapses and neurotransmitters so often emphasized. This perspective opens a vast, largely unexplored landscape of research and application. Geometry, once the language of altars and pyramids, now finds its most intricate canvas in the human nervous system. If we are to build machines that think, interfaces that feel, or therapies that restore, we must continue to think like Euclid and the Sulbakaras—believing that truth, and perhaps salvation, lies in the structure of space itself.